

- (a) Explain why the author believes that Franklin's experiment seemed about as wise as sticking a knife in a wall socket.

[2 marks]

- (b) Compare the properties of the string and the silk thread used in the experiment and suggest why Franklin used the string and the silk thread in the way that he did.

[2 marks]

- (c) A spark jumped from the key to Franklin's knuckle. How was the key able to acquire electrical energy to produce a spark in the first place and explain why the spark jumped to his knuckle?

[2 marks]

- (d) The author described the current in the string as a "relatively modest current". What factors would determine if the current was to be relatively modest or very large?

[2 marks]

- (e) What precautions should the Swedish physicist have taken to avoid becoming a 'fritter'?

[2 marks]

END OF TEST - TOTAL 60 MARKS

Answers

CHAPTER 1: MOTION AND FORCES

1. Chapter Questions

1.1

(a) speed is a scalar, velocity is a vector and involves direction.

(b) $t = 160 \text{ s}$

(i) distance = $400 + 200 \approx 600 \text{ m}$

displacement = 200 m

(ii) $\therefore$ average speed = $\frac{600 \text{ m}}{200 \text{ s}} = 3.00 \text{ ms}^{-1}$

(iii) instantaneous velocity is given by the slope of the graph at $t = 160 \text{ s}$. Slope is the same from $t = 100$ to $t = 220$ hence

$$\text{instantaneous vel} = \frac{-400 \text{ m}}{120 \text{ s}} = -3.33 \text{ ms}^{-1}$$

1.2.

(i) return journey

$$\text{average speed} = \frac{350 \text{ m}}{65 \text{ s}} = 5.4 \text{ ms}^{-1}$$

$$\text{average vel} = \frac{250 \text{ m}}{65 \text{ s}}$$

$$= 3.8 \text{ ms}^{-1} \text{ } 37^\circ \text{N of W}$$

(ii) the whole journey

$$\text{average speed} = \frac{700 \text{ m}}{130 \text{ s}} = 5.4 \text{ ms}^{-1}$$

$$\text{average vel} = 0 \text{ (since zero displacement)}$$

1.3.

Scalars – mass, density, volume, work done, speed, distance, area, temperature

Vectors – velocity, momentum, weight, acceleration, torque

1.4

The car is slowing down hence acceleration is negative.

1.5

The car is constantly changing direction hence its velocity (although not its speed) is changing. A changing velocity means acceleration is occurring.

1.6.

The stopping distance includes the distance travelled while the driver is reacting and beginning to apply the brakes.

1.7.

(a) Maximum displacement is

$$60 \text{ m} + 20 \text{ m} = 80 \text{ m from starting point}$$

(b) $t = 80 \text{ s}$, $s = 80 - (20 \times 2) = 40 \text{ m}$

1.8.

(a) Both vehicles are travelling at the same speed.

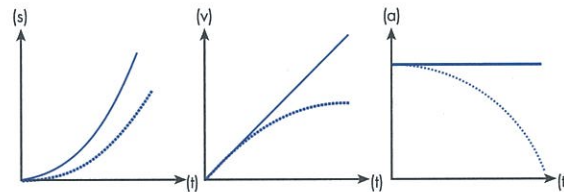
$$(b) a = \frac{v - u}{t} = \frac{30 - 0}{15} = 2.0 \text{ ms}^{-2}$$

(c) Areas under the two graphs are equal when $t = 22.5 \text{ s}$

1.9.

(a) Surface area, speed.

(b)



(i) s/t graph would be less steep and eventually straight

(ii) v/t graph – less steepness and eventually flat

(iii) a/t graph – starts at 9.8 ms^{-2} but eventually goes to zero.

1.10.

(a) $A \rightarrow B$ uniform acceleration ($\approx 9.8 \text{ ms}^{-2}$)

$B \rightarrow C$ acceleration becomes less and eventually zero

$C \rightarrow D$ constant velocity (no acc.)

$D \rightarrow E$ rapid uniform deceleration

$E \rightarrow F$ constant velocity

$F \rightarrow G$ rapid uniform deceleration

(b) Estimate area under graph

$$\approx [(30 \times 45) - (10 \times 20)] + [5 \times 95] \approx 1625$$

$$\therefore \approx 1600 \text{ m}$$

$$\text{Maximum acceleration} \approx \frac{5 - 45}{3}$$

$$\approx -13 \text{ ms}^{-2} \text{ (parachute opens)}$$

Height when parachute opened

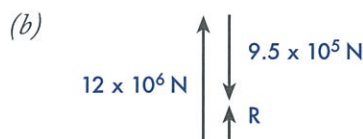
$\approx$ area under graph for $t \geq 30 \text{ s}$

$$\approx 5 \times 95 \approx 500 \text{ m}$$

$$\text{Terminal velocity} = 45 \text{ ms}^{-1} \text{ (parachute unopened)}$$



$R = -6000 \text{ N}$ (car is probably braking)
car accelerates to the left



$R = 2.5 \times 10^5 \text{ N}$ upwards
rocket is accelerating upwards



$R = (4 \times 10^5)^2 + (4 \times 10^5)^2$
 $= 5.66 \times 10^5 \text{ N}$ to the right
cargo boat accelerating to the right

1.12.

(a) (i) Your body tends to remain at rest (relative to the ground) while the car is accelerating forward. Relative to the car, you are moving backwards.

(b) (ii) Since the engine is turned off, the net force acting on the car is the frictional force opposing its motion.

1.13.

The negative (-) sign indicates the opposite direction to the sedan's initial velocity.

1.14.

(a) Impulse is the same since it is just the change in momentum.

(b) Force windscreen exerts on passenger equals $1.26 \times 10^4 \text{ N}$.

1.15.

(a) between $\approx t = (17.5 - 20.0) \text{ s}$
ie. greatest slope

(b) $F_{\text{max}} \approx 5 \text{ N}$

(c) $Ft = \text{area under graph} \approx 220 \text{ Ns}$.

1.16

(a) The hand pushes on the cricket ball (action) as it is being bowled and the cricket ball pushes back on the hand with an equal and opposite force (reaction). This means that relative to the hand the ball does not move although both hand and ball are being accelerated to a higher velocity by the bowler. When the hand no longer pushes on the ball the ball no longer accelerates but continues to move with whatever velocity it has.

(b) When you are in free fall the force acting on you is your weight or attraction due to the earth (action). You

are also attracting the earth towards you with an equal and opposite force (reaction). In fact the earth is also falling towards you (as you fall towards it) but at an almost insignificant rate due to its large mass.

(c) Your weight force acts down on the couch (action) while your couch pushes back up on you with an equal and opposite reaction force (reaction). This means there is no net force between you and the couch and you remain comfortably at rest.

1.17.

Although the forces are equal they are acting on different masses. The football has a smaller mass and accelerates the greatest amount. The foot, which is attached to the body, accelerates to a smaller extent and in the opposite direction (that is, it eventually stops).

1.18.

The stretched rubber balloon forces the air out as it returns to its normal size. Since overall momentum is maintained, the air rushing out in one direction (action), causes the balloon to move in the opposite direction (reaction).

1.19.

(a) $W = mg_e = (52)(9.8)$
 $= 510 \text{ N}$ (on earth)

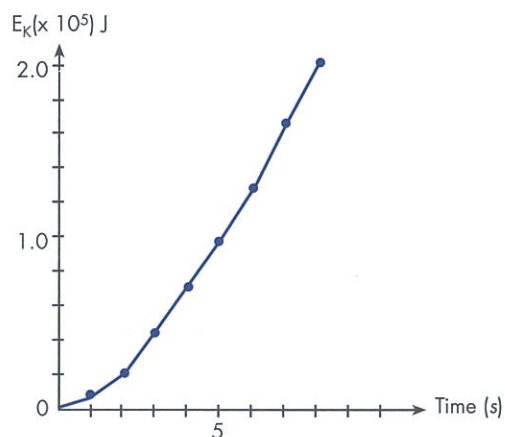
(b) $W = mg_m = (52)(1.6)$
 $= 83.2 \text{ N}$ (on the moon)

(c) $m = \frac{W}{9.8} = \frac{83.2}{9.8}$
 $= 8.50 \text{ kg}$

1.20.

(a) Values need to be calculated for graph.

$t \text{ (s)}$	$v \text{ (ms}^{-1}\text{)}$	$E_k \text{ (} \times 10^5 \text{ J)}$
1	3	0.045
2	6	0.18
3	9	0.405
4	12	0.72
5	14	0.98
6	16	1.28
7	18	1.62
8	20	2.00



(b)

(i) Power output at $t = 2.0 \text{ s}$

$$P = \frac{E_s}{t} = \frac{0.18 \times 10^5}{2} = 900 \text{ W}$$

(average to this point)

(ii) Power output at $t = 6.0 \text{ s}$

$$P = \frac{E_s}{t} = \frac{0.28 \times 10^5}{6} = 2.13 \text{ kW}$$

(average to this point)

Note: To find instantaneous power rating at a given time we would need to look at the slope of the E_k/t graph at that point.

$$\text{(eg. at } t=6, \text{ slope} \approx \frac{2 \times 10^5}{7} \approx 2.8 \times 10^4 \text{ W)}$$

(b)

average velocity = gradient

$$\begin{aligned} &= \frac{\Delta s}{\Delta t} \\ &= \frac{3.2 - 0.8}{4 - 1} \\ &= \frac{2.4}{3} \\ &= 0.8 \text{ cm s}^{-1} \end{aligned}$$

4.

(a) 0–2 s, vehicle is uniformly accelerating

2–5 s, vehicle has a constant velocity

5–7 s, vehicle is uniformly decelerating

7–9 s, vehicle has a constant velocity

(b) Displacement = area under the graph

$$\begin{aligned} &= \frac{1}{2} (2 \times 30) + (3 \times 30) + \frac{1}{2} (2 \times 10) \\ &\quad + (2 \times 20) + (2 \times 20) \\ &= 30 + 90 + 10 + 40 + 40 \\ &= 210 \text{ m} \end{aligned}$$

(c) Acceleration 0–2 s = gradient of graph

$$= \frac{30 - 0}{2 - 0} = 15 \text{ ms}^{-2}$$

Acceleration 5–7 s = gradient of graph

$$= \frac{20 - 30}{7 - 5} = -5 \text{ ms}^{-2}$$

From 0–2 s, the vehicle is accelerating at 15 ms^{-2} which represents a faster rate of change of velocity than for 5–7 s where the vehicle is decelerating at a slower rate of change of velocity.

1. Review Questions

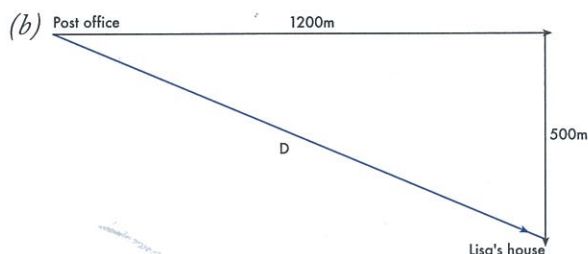
1.

(a) Displacement is the distance moved in a particular direction from a reference point. Distance, a scalar quantity, is independent of direction.

(b) Speed is the rate of change of distance while velocity is the rate of change of displacement (ie. the rate of change of distance in a given direction).

2. (a) Total distance

$$\begin{aligned} &= 600 \text{ m} + 500 \text{ m} + 400 \text{ m} + 400 \text{ m} + 600 \text{ m} \\ &= 2500 \text{ m} \end{aligned}$$



Displacement:

$$\begin{aligned} D^2 &= (1200)^2 + (500)^2 \\ &= 1.44 \times 10^6 + 2.5 \times 10^5 \end{aligned}$$

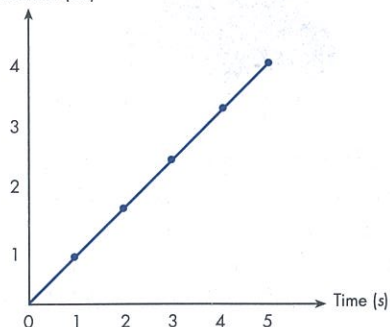
$$D^2 = 16.9 \times 10^5$$

$$\therefore D = 1300 \text{ m}$$

$$\therefore \text{Total displacement} = 1300 \text{ m}$$

3. Displacement (cm)

(a)



$$\begin{aligned} 5. \quad s &= ut + \frac{1}{2} at^2 \\ &= 0 + \frac{1}{2} g t^2 \\ &= \frac{1}{2} \times 9.8 \times (5)^2 \\ &= 122.5 \text{ m} \end{aligned}$$

$$\therefore \text{Depth of ravine} = 1.22 \times 10^2 \text{ m}$$

6.

(a) Let upwards be positive

$$v^2 = u^2 + 2gs$$

$$0 = u^2 + 2(-9.8)20$$

$$u^2 = 392$$

$$u = 19.8 \text{ ms}^{-1}$$

$$\text{Initial velocity} = 19.8 \text{ ms}^{-1}$$

(b)

(i) Time taken to reach max. height

$$v = u + at$$

$$0 = (+19.8) + (-9.8)t$$

$$(-9.8)t = -19.8$$

$$t = 2.02 \text{ s}$$

(ii) Time taken to fall 30 m

$$s = ut + \frac{1}{2} gt^2$$

$$30 = 0 + \frac{1}{2} \times 9.8 \times t^2$$

$$30 = 4.9t^2$$

$$t^2 = 6.12$$

$$\begin{aligned}
 t &= 2.47 \text{ s} \\
 \therefore \text{Time taken to reach the water} \\
 &= 2.02 \text{ s} + 2.47 \text{ s} \\
 &= 4.5 \text{ s}
 \end{aligned}$$

7. **Scalar quantity:** Specified by a number and units (if any) i.e. only has magnitude eg. temperature.
Vector quantity: As well as magnitude a vector quantity also has direction, eg. displacement.

8.

SCALAR	VECTOR
distance	acceleration
mass	force
volume	displacement
time	velocity

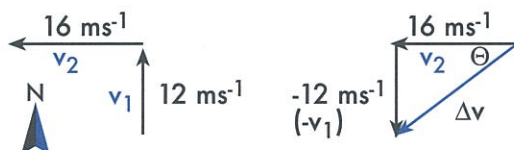
$$\begin{array}{c}
 15 \text{ N} \rightarrow \\
 + \\
 10 \text{ N} \rightarrow \\
 \hline
 = \rightarrow 25 \text{ N}
 \end{array}$$

$\therefore$ Force received by the young child and trolley is 25 N.

$$\begin{array}{c}
 15 \text{ N} \rightarrow \\
 - \\
 3 \text{ N} \leftarrow \\
 \hline
 = \rightarrow 12 \text{ N}
 \end{array}$$

$\therefore$ Force received by the young child and trolley is 12 N.

11.



$\therefore$ Change in velocity = 20 ms^{-1} , $W 36.9^\circ S$

12.

(a) $60 \text{ kmh}^{-1} \rightarrow$
 $2 \text{ kmh}^{-1} \rightarrow$

$$\begin{aligned}
 \text{John's velocity} &= 60 \text{ kmh}^{-1} + 2 \text{ kmh}^{-1} \\
 &= 62 \text{ kmh}^{-1} \text{ forwards}
 \end{aligned}$$

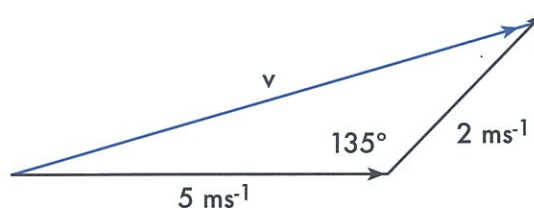
(b) $60 \text{ kmh}^{-1} \rightarrow$
 $2 \text{ kmh}^{-1} \leftarrow$

$$\begin{aligned}
 \text{John's velocity} &= 60 \text{ kmh}^{-1} + (-2 \text{ kmh}^{-1}) \\
 &= 58 \text{ kmh}^{-1} \text{ forwards}
 \end{aligned}$$

13.

$$\begin{aligned}
 v^2 &= (5^2) + (2^2) - (2)(5)(2)\cos 135^\circ \\
 &= 25 + 4 - 20(-0.7071) \\
 &= 43.14 \\
 \therefore v &= 6.6 \text{ ms}^{-1}
 \end{aligned}$$

$$\therefore \text{New velocity} = 6.6 \text{ ms}^{-1}$$



14.

$$\begin{aligned}
 \text{Horizontal Component} &= 20 \times \cos 30^\circ \\
 &= 17.3 \text{ N} \\
 \text{Vertical Component} &= 20 \times \sin 30^\circ \\
 &= 10 \text{ N}
 \end{aligned}$$

15.

- Cause moving objects to increase their acceleration.
- Change the direction in which objects are moving.
- Change the shape of objects.
- Oppose other forces acting on an object to prevent it from moving.

16.

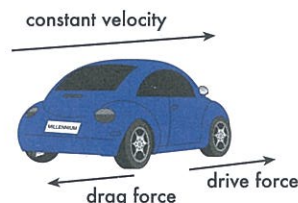
- The metal washer will fall into the glass jar.
- The force of friction between the cardboard and the glass jar acts for only a short period of time when the cardboard is quickly flicked by Mark's finger. The inertia of the metal washer on top of the fast moving cardboard restricts its motion, causing it to be left behind and fall into the glass jar.

17.

Every body continues in a state of rest or uniform motion in a straight line unless acted upon by some unbalanced external force.

18.

No. By applying pressure on the accelerator Mary is causing the car to provide a drive force to overcome external drag forces such as friction with the road and wind resistance. The net effect of the drive and drag forces is to enable the car to move at constant velocity since the total force acting on the car is zero.



19.

- The sudden braking of the train would normally have caused Nic to lean forward (Newton's First Law of Motion) resulting in him pushing the bottle past the glass.
- No. The inertia of the cool drink would have

carried some of it forward past the glass and onto the table as the train braked.

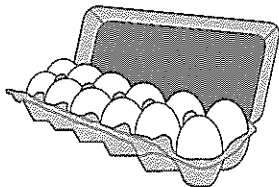
20.

The rate of change of momentum of a body is directly proportional to the external force acting on the body and occurs in the direction of the force.

21. No. Acceleration is caused by a force. Forces are not caused by accelerations. Therefore, the driver's stomach organs receive an acceleration due to a force acting on the driver.

22.

- (a) The inertia of the egg carton carried it forward during braking (Newton's First Law).
 (b) When the egg carton hit the floor, the rate of change of momentum of the eggs in the carton is decreased by the soft cardboard material. This longer time for the change in momentum produces a smaller force to act on the eggs - small enough not to break them.



23. If a body A exerts a force on a body B, then B exerts an oppositely directed force of the same size on A.

24. Susan. Since her starting blocks are attached to the ground then she will receive a reaction force in the direction she is running from the starting blocks as she pushes off against them (Newton's Third Law). Jacky's starting blocks will move off away from her as she pushes against them and provide little to no force to move forward (-she will most likely fall to the ground).

25.

As Tim made contact with the hard squash court floor while wearing the soft, thick rubber soled shoes, his feet and legs would receive less impulsive force. The softer shoe enables the change in momentum to occur over a longer period of time directing less force (and subsequently less pain) to his feet and legs.

26. The air bag increases the time taken for the driver to undergo a change of momentum. The driver is subjected to a smaller force which may otherwise have been sufficiently large (due to the more rapid change of momentum as the driver hits the steering wheel in the absence of the air bag) to cause serious injuries to the driver.

27. Let the direction of the 40 kg skater be positive.
 Momentum before the collision =
 Momentum after the collision

$$\begin{aligned} m_1 v_1 + m_2 v_2 &= (m_1 + m_2) v_3 \\ (40)(3) + (30)(-5) &= (40 + 30) v_3 \\ 120 - 150 &= 70 v_3 \\ -30 &= 70 v_3 \\ \therefore v &= -0.4 \end{aligned}$$

The common velocity is 0.4 ms^{-1} in the original direction of the 30 kg skater.

28. If Peta 'follows through' when hitting the ball her bat will be in contact with the ball for a longer period of time. This will increase the impulse resulting in the ball experiencing a larger change in momentum. Therefore, the 'follow through' increases the speed at which the ball leaves Peta's softball bat (increasing the chance of her hitting the ball past the nearby fielders).

$$\begin{aligned} Ft &= mv - mu \\ F \times 0.1 &= (0.1)(0) - (0.1)(10) \\ 0.1F &= 0 - 1 \\ \therefore F &= -10\text{N} \end{aligned}$$

$\therefore$ Force = 10 N in the opposite direction to which the ball was moving.

30.

(a) resultant force

$$\begin{aligned} F &= \text{applied force} - \text{retarding force} \\ &= 15\text{ N} - 3\text{ N} = 12\text{ N} \\ F &= ma \\ 12 &= 60a \\ \therefore a &= 0.2 \text{ ms}^{-2} \end{aligned}$$

(b) $F = \frac{\Delta p}{\Delta t}$

$$\begin{aligned} \Delta p &= F \times \Delta t \\ &= 12 \times 4 \\ &= 48 \text{ kgms}^{-1} \end{aligned}$$

$$\therefore \text{Change in momentum} = 48 \text{ kgms}^{-1}$$

(c) $s = ut + \frac{1}{2}at^2$

$$\begin{aligned} &= 0 + \frac{1}{2}(0.2)(4)^2 \\ &= 1.6 \text{ m} \\ \therefore \text{Distance travelled} &= 1.6 \text{ m} \end{aligned}$$

31.

- (a) The mass of an object is a measure of the matter in it.
 (b) The inertia of an object is its reluctance to resist changes in its motion (ie. its reluctance to start moving, or to stop moving once it has started).

32.

(a) $F = mg$

$$\begin{aligned} &= 1500 \times 9.8 \\ &= 1.47 \times 10^4 \text{ N} \end{aligned}$$

$$\begin{aligned}
 (b) \quad W &= Fs \\
 &= 1.47 \times 10^4 \times 3 \\
 &= 4.4 \times 10^4 \text{ J}
 \end{aligned}$$

$$33. \quad 60 \text{ kmh}^{-1} = 16.7 \text{ ms}^{-1}$$

$$\begin{aligned}
 (a) \quad E_K &= \frac{1}{2} mv^2 \\
 &= (\frac{1}{2})(1500)(16.7)^2 \\
 &= 2.1 \times 10^5 \text{ J}
 \end{aligned}$$

$$\begin{aligned}
 (b) \quad v^2 &= u^2 + 2as \\
 0 &= (16.7)^2 + (2)(a)(120) \\
 240a &= -278.9 \\
 a &= -1.16 \text{ ms}^{-2} \\
 F &= ma \\
 &= (1500)(-1.16) \\
 &= 1.74 \times 10^3 \text{ N}
 \end{aligned}$$

$\therefore$ Stopping force = $1.74 \times 10^3 \text{ N}$, in the opposite direction to which the car was travelling.

$$\begin{aligned}
 (c) \quad v &= u + at \\
 0 &= 16.7 + (-1.16)t
 \end{aligned}$$

$$\therefore 1.16t = 16.7$$

$$\therefore t = 14.4 \text{ s}$$

$$\therefore \text{Time taken to stop} = 14.4 \text{ s}$$

$$\begin{aligned}
 (d) \quad \text{Work done to stop} &= \text{loss in kinetic energy} \\
 &= 2.1 \times 10^5 \text{ J} \\
 \text{OR } W &= Fs \\
 &= 1.74 \times 10^3 \times 120 \\
 &= 2.1 \times 10^5 \text{ J}
 \end{aligned}$$

$$34. \quad 100 \text{ kmh}^{-1} = 27.8 \text{ ms}^{-1}$$

$$\begin{aligned}
 E_K \text{ of car} &= \frac{1}{2} mv^2 \\
 &= (\frac{1}{2})(1600)(27.8)^2 \\
 &= 6.17 \times 10^5 \text{ J}
 \end{aligned}$$

$$\begin{aligned}
 E_K \text{ of truck} &= E_K \text{ of car} = \frac{1}{2} mv^2 \\
 \therefore 6.17 \times 10^5 &= (\frac{1}{2})(8000)v^2 \\
 \therefore v^2 &= 154.3 \\
 \therefore v &= 12.4 \text{ ms}^{-1} \\
 \therefore \text{Velocity of truck} &= 12.4 \text{ ms}^{-1} (44.6 \text{ kmh}^{-1})
 \end{aligned}$$

$$35. \quad E_K \text{ of car} = E_p \text{ of spring}$$

$$\begin{aligned}
 \frac{1}{2} mv^2 &= 2 \\
 (\frac{1}{2})(0.1)v^2 &= 2 \\
 v^2 &= 40 \\
 v &= 6.3 \text{ ms}^{-1}
 \end{aligned}$$

$$\begin{aligned}
 36. \quad E_K \text{ lost by dart} &= E_p \text{ of block + dart} \\
 \frac{1}{2} m_d v^2 &= (m_b + m_d)gh \\
 (\frac{1}{2})(0.05)v^2 &= (0.50+0.05)(9.8)(0.040) \\
 \therefore \text{velocity of block and dart after impact} &= 2.94 \text{ ms}^{-1}
 \end{aligned}$$

$$\begin{aligned}
 \text{Also } P_i(\text{dart}) &= P_f(\text{dart + block}) \\
 (0.050)(v) &= (0.50+0.05)(2.94) \\
 \therefore \text{Initial vel dart} &= 32.3 \text{ ms}^{-1}
 \end{aligned}$$

37.

$$\begin{aligned}
 (a) \quad \text{Alan falls through a height of } 30 \text{ cm} - 15 \text{ cm} \\
 &= 15 \text{ cm}
 \end{aligned}$$

$$\begin{aligned}
 \therefore \text{Alan's energy before impact} &= \text{loss in potential energy} \\
 &= mgh \\
 &= (45)(9.8)(0.15) \\
 &= 66.2 \text{ J}
 \end{aligned}$$

$$\begin{aligned}
 (b) \quad E_K \text{ gained} &= E_p \text{ lost} \\
 \frac{1}{2} mv^2 &= mgh \\
 (\frac{1}{2})(45)v^2 &= 66.2 \\
 v^2 &= 2.94
 \end{aligned}$$

$$\therefore v = 1.72 \text{ ms}^{-1}$$

Alan's velocity just before impact = 1.72 ms^{-1} downwards

$$\begin{aligned}
 (c) \quad \text{Momentum before} &= \text{Momentum after} \\
 (mu)_{\text{Alan}} + (mu)_{\text{peg}} &= (mv)_{\text{Alan}} + (mv)_{\text{peg}}
 \end{aligned}$$

$$(45)(1.72) + (0.20)(0) = 45v + 0.2v$$

$$\text{where } v = v_{\text{Alan}} = v_{\text{peg}}$$

$$77.4 = 45.2v$$

$$v = 1.71 \text{ ms}^{-1}$$

$$\therefore \text{velocity of the peg after impact} = 1.71 \text{ ms}^{-1}$$

$$(d) \quad \text{Energy before impact} = 66.2 \text{ J}$$

$$\text{Energy after impact}$$

$$\begin{aligned}
 &= E_{K\text{Alan}} + E_{K\text{peg}} \\
 &= \frac{1}{2} mv^2 + \frac{1}{2} mv^2 \\
 &= (\frac{1}{2})(45)(1.71)^2 + (\frac{1}{2})(0.2)(1.71)^2 \\
 &= 65.8 + 0.29 \\
 &= 66.1 \text{ J}
 \end{aligned}$$

$$\therefore \text{Energy loss due to impact} = 0.1 \text{ J}$$

$$(e) \quad \text{Work done} = E \text{ lost}$$

$$E \text{ lost} = \text{Initial } E_K \text{ of Alan} - E_K \text{ of Alan on peg that is } 0.6 \text{ cm high}$$

$$= 66.2 - mgh$$

$$= 66.2 - 45 \times 9.8 \times 0.06$$

$$E \text{ lost} = 39.7 \text{ J} = \text{Work done}$$

$$F = \frac{W}{s} = \frac{39.7}{0.09} = 442 \text{ N}$$

38.

$$(a) \quad \text{Vertical height climbed by Shirley}$$

$$= 30 \times 0.25$$

$$= 7.5 \text{ m}$$

$$\text{Shirley's weight} = mg$$

$$= 50 \times 9.8$$

$$= 490 \text{ N}$$

$$\therefore \text{Work done by Shirley} = \text{force (to lift her weight)} \times \text{distance (vertical height)}$$

$$= 490 \times 7.5$$

$$= 3.68 \times 10^3 \text{ J}$$

$$(b) \quad \text{Power output} = \frac{\text{work done}}{\text{time taken}}$$

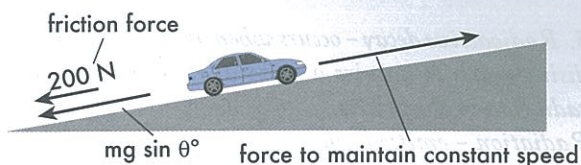
$$= \frac{3.68 \times 10^3}{10}$$

$$= 3.68 \times 10^2 \text{ W}$$

$$= 3.68 \times 10^2 \text{ W}$$

$$\therefore \text{Shirley's power output} = 3.68 \times 10^2 \text{ W}$$

39.



$$P = \frac{W}{t} = F \times \frac{s}{t} = Fv$$

Force to maintain constant speed =
Force to overcome gravity along the slope +
Force to overcome friction

$$\begin{aligned} &= mg \sin 10 + 200 \\ &= 1500 \times 9.8 \times \sin 10 + 200 \\ &= 2.75 \times 10^3 \text{ N} \end{aligned}$$

$$\begin{aligned} \text{Power developed} &= F \times v \\ &= 2.75 \times 10^3 \times 15 \\ &= 4.13 \times 10^4 \text{ W (41.3 kW)} \end{aligned}$$

40.

$$\begin{aligned} \text{(a) Work done} &= \text{area under the graph} \\ &= (5 \times 0.5) + [(5 \times 0.25) + \frac{1}{2} (5 \times 0.25)] + (10 \times 0.75) \\ &= 2.5 + 1.88 + 7.5 \\ &= 11.9 \text{ J} \end{aligned}$$

$$\begin{aligned} \text{(b) Average power exerted} &= \frac{\text{work done}}{\text{time}} \\ &= \frac{11.9}{10} = 1.19 \text{ W} \end{aligned}$$

CHAPTER 2: NUCLEAR PHYSICS

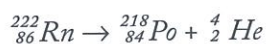
2. Chapter Questions

2.1.

- (a) ${}_{90}^{232}\text{C} \rightarrow {}_{88}^{228}\text{Ra} + {}_2^4\text{He}$, alpha particle
(b) ${}_{12}^{24}\text{Mg}^* \rightarrow {}_{12}^{24}\text{Mg} + \gamma$, gamma ray
(c) ${}_1^3\text{H} \rightarrow {}_2^3\text{He} + {}_{-1}^0\text{e}$, electron (β^-)
(d) ${}_6^{11}\text{C} \rightarrow {}_5^{11}\text{B} + {}_{+1}^0\text{e}$, positron (β^+)

2.2.

- (a) 25% original nuclei remain
 $\therefore$ 2 half-lives have elapsed
 $\therefore$ 1 half-life = $7.6 / 2 = 3.8$ days
(b) Nuclei reduced by $(1/2)$ for each half-life.
To have less than 1% $(1/2)^n \leq 1/100$
We know $(1/2)^6 = 1/64$ and $(1/2)^7 = 1/132$
 $\therefore$ require a little less than 7 half-lives
(7×3.8 days) ≈ 25 days
(c) Radium 226
(d) Polonium 218



2.3.

- (a) Specimens older than say 50,000 years will have very little carbon-14 activity (less than 0.2%). This makes precise measurements difficult. Specimens that are say only 100 years old will have a carbon-14 to carbon-12 ratio which is only very slightly ($< 1\%$) different to modern day specimens. Again, errors in measurements will make accurate datings impossible.
(b) • Carbon-14 concentration in the atmosphere was the same in the past as it is today.
• The rate of decay of carbon-14 has always been constant.

2.4.

- (a) Thorium, Radium, Thallium, Lead, etc.
(b) Uranium-238 or lead-206 depending on the age of the rock. U-238 has a very long half-life and probably more than half remains. If the rock was more than one half-life of age, the majority of all the other daughter isotopes would have all decayed to Pb-206, since they have relatively short half-lives.
(c) Radon-222 or polonium-214. Radon is a gas with a short half-life. Po-214 has an extremely short half-life.
(d) Approximately 4.5 billion years. Uranium-238 eventually decays to lead-206. Most of the isotopes in the decay series have relatively short half-lives. Hence if half the U-238 has decayed one half-life has elapsed.

2.5.

- high activity - high number of decays per second - this ensures easier detection
- physical half-life - time taken for half of the radioisotope to decay. Short half-life means the patient is not exposed for long periods.
- biological half-life - time taken for body to expel half of the radioisotope by normal bodily functions. Short biological half-life also means minimum exposure to patient.

2.6.

- (a) Absorbed dose simply considers radiation energy absorbed by the body. Dose equivalent factors in the different biological effects of α , β , γ radiation.
(b) Dose equivalent measured in sievert gives more accurate measure of biological effect.

2.7.

- (a) LD50/30 means that 50% of exposed people would die within 30 days.
(b) absorbed dose = $\frac{\text{energy absorbed}}{\text{mass of tissue}}$

$$\begin{aligned} \therefore \text{energy absorbed} &= (4) (85) \\ &= 340 \text{ J} \\ &(\text{since } QF \text{ for } \gamma \text{ radiation} = 1) \end{aligned}$$

- 2.8.
- (a) The earth's atmosphere acts as a shield and absorbs cosmic radiation.
- (b) Pilots and air crew are exposed for a much greater time during their working life. A record of total exposure would ensure that a "safe limit" was not surpassed.
- (c) $\text{dose} = \text{rate} \times \text{time}$
 $= (5 \text{ uSv}) (200 \text{ h}) = 1000 \text{ uSv}$
 yearly dose at ground level
 $= (0.03) (24) (365)$
 $\approx 263 \text{ uSv}$
 Pilot receives the equivalent of about four yearly doses during his flights.

- 2.9.
- Graph shows highest BE/nucleon for medium sized atoms. Light atoms can combine to a more stable state and release binding energy. Large atoms can do so by splitting.

- 2.10.
- for $A = 200$ BE/nucleon $\approx 7.7 \text{ MeV/nucleon}$
 for $A = 100$ BE/nucleon $\approx 8.2 \text{ MeV/nucleon}$
 Energy released $\approx (200) (0.5) \approx 100 \text{ MeV}$

- 2.11. Energy released $= (0.21534) (931.5)$
 $= 200 \text{ MeV}$

2. Review Questions

2.1 Radioactivity

- 1.
- (a) Atomic number: the number of protons in the nucleus of all atoms of a particular element.
- (b) Mass number: the number of protons and neutrons in the nucleus of an atom.
- (c) Isotope: isotopes are atoms which have the same proton number but different number of neutrons.
- (d) Nuclide: refers to an atom with a particular number of protons and a particular number of neutrons.
- (e) Atomic mass unit: the unit of atomic mass is equal to $1/12$ of the mass of the $^{12}_6\text{C}$ atom.

2. (a) 6,6
 (b) 1,0
 (c) 5,6
 (d) 92,143

3. (a) $^{23}_{11}\text{Na}$
 (b) $^{16}_8\text{O}$
 (c) $^{39}_{19}\text{K}$
 (d) $^{90}_{38}\text{Sr}$

4. The nuclei have too few or too many neutrons which cause instability and radioactive decay.

5. Radioactive decay – occurs when an unstable nucleus disintegrates to acquire a more stable state and emits radiation in the process.

Radiation – emitted when an unstable nucleus decays and may be energetic particles (alpha particle, beta particle) or energy in the form of electromagnetic radiation with a very short wavelength (gamma rays).

6. Alpha particle – consists of two protons and two neutrons (identical to a helium nucleus).

Beta particle – electron.

Gamma radiation – electromagnetic radiation of very short wavelength.

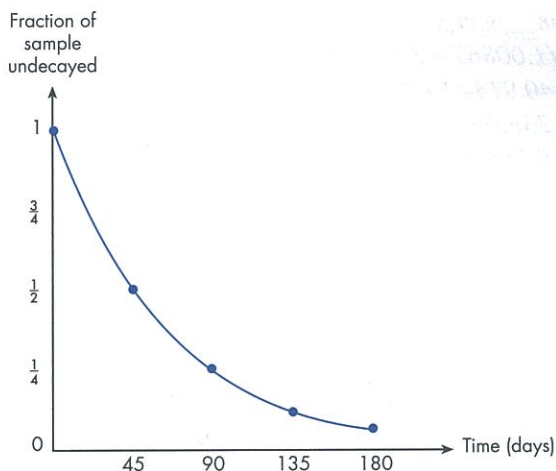
- 7.
- | | | |
|------------------------|-----------|-----------|
| (a) alpha | (b) beta | (c) alpha |
| (d) beta | (e) gamma | (f) beta |
| (g) gamma | (h) alpha | (i) gamma |
| (j) alpha, beta, gamma | | (k) beta |

- 8.
- (a)
- (i) The thin piece of paper would absorb alpha particles, only some beta particles and no gamma radiation. The slight drop in count rate suggests the source does not emit alpha particles.
- (ii) The magnetic field deflects some radiation (eg. beta) away from the detector and not other (eg. gamma).
- (iii) Gamma rays are not deflected by a magnetic field nor the sheet of aluminium – the count rate should be the same in both cases. Beta particles are absorbed by this thickness of aluminium.
- (b) Beta and gamma radiation.

- 9.
- (a) $^A_Z\text{X} \rightarrow ^{A-4}_{Z-2}\text{Y} + ^4_2\text{He}$ (b) $^A_Z\text{X} \rightarrow ^A_{Z+1}\text{Y} + ^0_{-1}\beta$

10. (a) ^4_2He (b) $^{90}_{39}\text{Y}$ (c) $^{14}_6\text{C}$ (d) $^{226}_{86}\text{Rn}$

- 11.
- 50% of the atoms decay over 45 days.
 $\therefore$ half-life of = 45 days.
 $1/2$ of the sample remains undecayed after 45 days
 $1/4$ of the sample remains undecayed after 90 days
 $1/8$ of the sample remains undecayed after 135 days
 $1/16$ of the sample remains undecayed after 180 days
 $\therefore 1/16$ of the sample remains undecayed after 6 months.



12. Half life of the sample is 3 minutes.

Time (mins)	0	3.00	6.00	9.00	12.0	15.0
Fraction of Activity	1	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{8}$	$\frac{1}{16}$	$\frac{1}{32}$

Time (mins)	18.0	21.0	24.0	27.0	30.0
Fraction of Activity	$\frac{1}{64}$	$\frac{1}{128}$	$\frac{1}{256}$	$\frac{1}{512}$	$\frac{1}{1024}$

After 30.0 minutes the activity will be reduced to $\frac{1}{2}$ of its original rate.

$$\therefore \text{Activity } \frac{1}{1024} \times 4.30 \times 10^4 \\ = 42.0 \text{ decays per second.}$$

13. Many radionuclides with a short half-life would have an unsuitably low dose rate by the time they arrived in Australia and then were distributed to the appropriate hospitals.

14. The radionuclide concentrates in the organ to which the imaging is to apply. Successful imaging will rely on sufficient radionuclide to be concentrated in the organ and for the radionuclide to have enough activity. To prevent excessive radiation exposure to nearby and other body tissue a radionuclide with a short half-life is used i.e. its activity will quickly fall away to a very low level.

2.2 Radiation – Effects and uses

15. The localisation and concentration of the radioactive iodine results in the surrounding thyroid cancer receiving a much higher radiation dose than other body tissues. This higher radiation dose is sufficient to destroy the thyroid cancer cells.

16.

- (i) detect defects in welds, alloys, pipes
- (ii) food preservation
- (iii) sterilisation of insect pests

(iv) sterilisation of medical dressings

17.

- (a) (i) watches with luminous dials (0.01 millisievert/annum)
- (ii) colour T.V. sets ($< 0.01 \text{ mSv/annum}$)
- (iii) ceramics/glass tinted with thorium salts
- (b) The exposures are considered to be trivial and are not dangerous.

18. The radiation dose is higher because the subsonic airliner takes longer to complete the trip. The crew member receives a lower radiation dose at the lower altitude but the time of exposure is longer due to the longer flight time.

19.

- (a) ${}_{90}^{232}\text{Th} \rightarrow {}_{88}^{228}\text{Ra} + {}_2^4\text{He}$
- (b) Alpha particles are more likely to penetrate the softer/thinner internal tissues than the tougher external skin layer – such as on the arms, hands and legs.
- (c) Do not eat/smoke in dust prone areas, thoroughly wash hands before eating.

20.

- (i) cosmic radiation
- (ii) radionuclides in the soil/rocks
- (iii) radioactivity of our body tissues (containing potassium-40)
- (iv) radon – in houses

21. alpha, alpha, beta

22.

- (a) The radon-222 gas can be inhaled, its decay products being deposited on the linings of both the air passages and the lungs. Some decay products have a long half-life and irradiate the tissue linings for many years.
- (b) (i) Provision of adequate ventilation.
- (ii) Sealing porous building materials.

23. Ionising radiation may

- (i) kill a body cell, preventing further cell division
- (ii) damage a cell which divides to produce abnormalities in the daughter cells.

24. Alpha particle – it can only penetrate a small section of the body tissue before being absorbed. It therefore rapidly loses its energy causing ionisations to occur in cells which are close together.

25.

- (i) keep an adequate distance from the source
- (ii) use shielding techniques.

2.3 Nuclear Energy

26. (a) ${}_{92}^{238}\text{U}$

$$\begin{aligned}\text{mass of protons} &= 92 \times 1.00728 \text{ u} \\ &= 92.66976 \text{ u} \\ \text{mass of neutrons} &= 146 \times 1.00867 \text{ u} \\ &= 147.26582 \text{ u} \\ \text{mass of electrons} &= 92 \times 0.00055 \text{ u} \\ &= 0.05060 \text{ u} \\ \text{Total mass} &= 239.98618 \text{ u}\end{aligned}$$

The sum of the mass of the individual particles is greater than their combined mass within the atom.

(b) When the atom is formed some of the mass of the constituent particles is converted into energy ($E = mc^2$) leaving the mass of the atom less than the sum of the masses of its constituents.

(c) Mass defect.

27.

$$\begin{aligned}\text{Mass defect} &= 239.98618 \text{ u} - 238.05080 \text{ u} \\ &= 1.93538 \text{ u}\end{aligned}$$

(a) Convert u to kg where

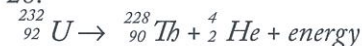
$$\begin{aligned}1 \text{ u} &= 1.660566 \times 10^{-27} \text{ kg} \\ 1.93538 \text{ u} &= 1.93538 \times 1.660566 \times 10^{-27}\end{aligned}$$

$$\begin{aligned}E &= mc^2 \\ &= 3.21382 \times 10^{-27} \times (3.00 \times 10^8)^2 \\ \therefore \text{Binding energy} &= 2.892 \times 10^{-10} \text{ J} \\ \text{Binding energy per nucleon} \\ &= 2.892 \times 10^{-10} \div 238 \\ &= 1.215 \times 10^{-12} \text{ J}\end{aligned}$$

(b) Binding energy

$$\begin{aligned}&= 2.892 \times 10^{-10} \div 1.602 \times 10^{-19} \\ &= 1.805 \times 10^9 \text{ eV} \\ \text{Binding energy per nucleon} \\ &= 1.805 \times 10^9 \div 238 \\ &= 7.59 \text{ MeV}\end{aligned}$$

28.

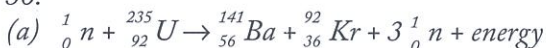


$\therefore$ Mass difference

$$\begin{aligned}&= m_{\text{LHS}} - m_{\text{RHS}} \\ &= 232.037131 \text{ u} - (228.028716 \text{ u} + 4.002602 \text{ u}) \\ &= 0.005813 \text{ u} \\ &= 9.6529 \times 10^{-30} \text{ kg} \\ \text{Energy released} &= mc^2 \\ &= 9.6612 \times 10^{-30} \times (3.00 \times 10^8)^2 \\ &= 8.6876 \times 10^{-13} \text{ J} \\ &= (8.6876 \times 10^{-13}) \div (1.602 \times 10^{-19}) \\ &= 5.42 \text{ MeV}\end{aligned}$$

29. A moderator, usually made of graphite, slows down the neutrons and increases the probability of neutron capture.

30.



(b) Mass difference

$$\begin{aligned}&= m_{\text{LHS}} - m_{\text{RHS}} \\ &= (1.00867 + 235.04392) - \\ &\quad (140.91440 + 91.92630 + 3.02601) \\ &= 236.05259 - 235.86671 \\ &= 0.18588 \text{ u} \\ &= 3.08666 \times 10^{-28} \text{ kg}\end{aligned}$$

Energy released

$$\begin{aligned}&= mc^2 \\ &= 3.08666 \times 10^{-28} \times (3.00 \times 10^8)^2 \\ &= 2.77799 \times 10^{-11} \text{ J} \\ &= (2.77799 \times 10^{-11}) \div (1.602 \times 10^{-19}) \\ &= 1.73407 \times 10^8 \text{ eV} \\ \therefore \text{Energy released} &= 173 \text{ MeV}\end{aligned}$$

31.

From Q.30, energy released by 1 atom

$$\begin{aligned}&= 2.78 \times 10^{-11} \text{ J} \\ \text{With 40\% efficiency, energy released} \\ &= 0.4 \times 2.78 \times 10^{-11} \\ &= 1.11 \times 10^{-11} \text{ J}\end{aligned}$$

No. of atoms required to release $300 \times 10^6 \text{ J}$ in 1 sec

$$= \frac{300 \times 10^6}{1.11 \times 10^{-11}}$$

$$= 2.70 \times 10^{19} \text{ atoms per sec}$$

No. of atoms fissioned in 1 year

$$\begin{aligned}&= 2.70 \times 10^{19} \times (365 \times 24 \times 60 \times 60) \\ &= 8.51 \times 10^{26} \text{ atoms}\end{aligned}$$

Now, 235.04392 g of contains 6.02×10^{23} atoms

$\therefore$ mass required

$$= \frac{8.51 \times 10^{26}}{6.02 \times 10^{23}} \times 235.04392$$

$$= 3.32 \times 10^5 \text{ g}$$

$$\text{Mass of } {}_{92}^{232}\text{U required} = 332 \text{ kg per year}$$

CHAPTER 3: HEATING AND COOLING

3. Chapter Questions

3.1

Heat refers to energy that is transferred from one body to another. A body does not actually contain heat but contains internal energy.

3.2

(a) (i) cold water

(ii) hot nail

(b) The average kinetic energy of the atoms of metal will fall while that of the molecules of water will rise until they are the same.

3.3

(a) The ocean contains the greatest amount of internal energy due to its vast size.

(b) On entering the water your body (at 37°C) transfers heat to the cooler ocean water and feels cold. Even though the ocean has in total more internal energy the average kinetic energy of the water particles is less than those of your body. Hence heat is continually transferred from you to the ocean water.

3.4

(a) The hand from the icy water would feel warm whereas that from the warm water would feel cold.

(b) The skin of the hand in the icy water would have reached a temperature close to zero while held there. When placed in the tap water which is at a comparatively higher temperature heat would flow to the hand giving the sensation of warmth. The reverse occurs to the hand going from warm to cold.

3.5

The fans keep air moving past your skin. This has a cooling effect as it aids the evaporation of moisture from your body by continually removing air saturated with water vapour from your sweat.

3.6

- Wet clothes will dry out even on a cold day.
- Water spilt on the floor will eventually disappear (evaporate), even on cold days.

3.7

Wind increases the evaporation of water off a wet body. This evaporation draws a lot of heat from the body (latent heat of vaporisation) and will eventually lead to hypothermia.

3.8

Evaporative air conditioners depend on the evaporation of water to create a cooling effect. On humid days evaporation is more difficult to achieve and hence they are not as effective.

3.9.

The room will not change in temperature. The refrigerator is simply a heat pump, drawing heat from inside the fridge and letting it out behind the fridge. If the fridge door is open there is no net effect. In fact the temperature of the room may increase because the energy used by the motor and pump will be converted to heat energy.

3.10

(a) Heat from the fire reaches your face and hands by radiation. Other parts of your body not facing the fire are shielded from radiation and feel cool.

(b) Hot air near and above the burning logs is less dense than cold air and rises. As it rises it causes a vertical flickering of the flames.

(c) Wood is a very poor conductor of heat. One end of

the stick can be quite cold while the other end is actually burning.

3.11

Air is an excellent insulator. Also trapped air is not able to move and transfer heat by convection.

3.12

(a) The earth is at a much lower temperature than the sun. Cooler bodies radiate longer wavelengths.

(b) The earth's temperature would be much lower since much of the heat that the atmosphere now re-radiates back to earth would escape into space. In effect the greenhouse effect of the atmosphere would be reduced.

3.13

In the late afternoon the sun's rays reach us more obliquely. This means that they pass through more atmosphere, hence more absorption, and they cover a larger surface area in the ground, hence less intensity.

3. Review Questions

1. (a) Heat is a form of energy which is transferred from one place to another due to a difference in temperature between them.

(b) Internal energy refers to the sum of the kinetic and potential energies of the molecules in an object.

(c) The temperature of an object is its degree of hotness (or coldness) on a chosen temperature scale. Temperature is a measure of the average kinetic energy of the individual molecules in the object.

2. The piece of lead possesses internal energy not heat. Heat is energy transferred from the lead as a result of change in temperature (eg. a piece of lead heated by a Bunsen burner then placed in a beaker of cold water - heat is transferred from the lead to the water because of their difference in temperature).

3. No. Correctly speaking, the term 'internal energy' should have been used.

4. No. The internal energies of the lead and water may be different but both the water and the lead are at the same temperature. Heat will only be transferred from one to the other if they each have a different temperature.

5. Electrical resistance, voltage generated at junction of two metals, expansion.

6. (a) The mercury-in-glass thermometer uses the property that the length of a column of mercury changes with a change in temperature.

(b) The platinum resistance thermometer uses the property that the electrical resistance of platinum changes with a change in temperature.

7.

- (a) (i) ice point, 0°C ; steam point, 100°C
 (ii) absolute zero, 0 K ; ice point, 273 K ,
 steam point, 373 K

- (b) ice point – the temperature of melting pure ice
 steam point – the temperature at which pure water can exist in equilibrium with its vapour (at standard atmospheric pressure)

8.

- (a) (i) 310 K (ii) 253 K (iii) 387 K
 (b) (i) -93°C (ii) 20°C (iii) 77°C

9. The ice must be pure. Impurities will lower the freezing point of water.

10.

- (a) They were both at the same temperature.
 (b) The jam has a higher specific heat capacity than the pastry and can transfer a larger amount of heat to the tongue, compared to pastry, when it is tasted. (It is the water content of the jam which is responsible for the higher specific heat – and burning Cathy's tongue!)

11. Paraffin oil, as it is most likely to have a smaller specific heat capacity than water.

12. As a coolant due to its very high specific heat capacity.

13. $m = 1.50\text{ kg}$
 $T_1 = 25.0^{\circ}\text{C}$
 $Q = 3.00 \times 10^5\text{ J}$
 $c = 4.18 \times 10^3\text{ J kg}^{-1}\text{C}^{-1}$
 $T_2 = ?$

$$Q = mc\Delta T$$

$$3.00 \times 10^5 = 1.50 \times 4.18 \times 10^3 \times \Delta T$$

$$\Delta T = \frac{3.00 \times 10^5}{1.50 \times 4.18 \times 10^3}$$

$$\therefore \Delta T = 47.8^{\circ}\text{C}$$

$$\therefore \text{Temperature after 2 mins} = 47.8 + 25.0$$

$$= 72.8^{\circ}\text{C}$$

14. $m = 150\text{ kg}$
 $T_1 = 20.0^{\circ}\text{C}$
 $T_2 = 65.0^{\circ}\text{C}$
 $c = 4.18 \times 10^3\text{ J kg}^{-1}\text{C}^{-1}$
 $Q = ?$

(a) $Q = mc\Delta T$
 $= 150 \times 4.18 \times 10^3 \times 45.0$
 $= 28.2 \times 10^6\text{ J}$
 $(= 28.2\text{ MJ})$

(b) $\text{Time} = \frac{2.82 \times 10^7}{7.00 \times 10^3}$
 $= 4031\text{ s}$
 $= 67.2\text{ minutes}$

15.

$$T_{1\text{ iron}} = 90.0^{\circ}\text{C}$$

$$T_{2\text{ iron}} = T$$

$$T_{1\text{ water}} = 20.0^{\circ}\text{C}$$

$$T_{2\text{ water}} = T$$

$$m_{\text{water}} = 30.0\text{ g} = 0.030\text{ kg}$$

$$m_{\text{iron}} = 50.0\text{ g} = 0.050\text{ kg}$$

$$c_{\text{iron}} = 4.60 \times 10^2\text{ J kg}^{-1}\text{C}^{-1}$$

$$c_{\text{water}} = 4.18 \times 10^3\text{ J kg}^{-1}\text{C}^{-1}$$

$$\text{Heat lost by iron} = \text{Heat gained by water}$$

$$m c \Delta T_{\text{iron}} = m c \Delta T_{\text{water}}$$

$$0.050 \times 4.60 \times 10^2 \times (90.0 - T) =$$

$$0.030 \times 4.18 \times 10^3 \times (T - 20.0)$$

$$2070 - 23T = 125.4T - 2508$$

$$148.4T = 4578$$

$$T = 30.8^{\circ}\text{C}$$

$$\therefore \text{Final temperature} = 30.8^{\circ}\text{C}$$

16.

In addition to No. 15

$$T_{1\text{ alum}} = 20.0^{\circ}\text{C}$$

$$T_{2\text{ alum}} = T$$

$$m_{\text{alum}} = 0.200\text{ kg}$$

$$c_{\text{alum}} = 9.00 \times 10^2\text{ J kg}^{-1}\text{C}^{-1}$$

$$\text{Heat lost by iron} = \text{Heat gained by water} +$$

$$\text{heat gained by aluminium}$$

$$m c \Delta T_{\text{iron}} = m c \Delta T_{\text{water}} + m c \Delta T_{\text{alum}}$$

$$0.050 \times 4.60 \times 10^2 \times (90.0 - T) =$$

$$0.030 \times 4.18 \times 10^3 \times (T - 20.0) +$$

$$0.200 \times 9.00 \times 10^2 \times (T - 20.0)$$

$$2070 - 23T = 125.4T - 2508 + 180T$$

$$-3600$$

$$328.4T = 8178$$

$$T = 24.9^{\circ}\text{C}$$

$$\therefore \text{Final temperature} = 24.9^{\circ}\text{C}$$

17. The heat is used to overcome attractive forces between the molecules in the solid. This loosens the molecules from their 'fixed' positions to allow them to 'flow' over each other.

18.

$$m_{\text{kettle}} = 3.00\text{ kg}$$

$$c_{\text{kettle}} = 4.00 \times 10^2\text{ J kg}^{-1}\text{C}^{-1}$$

$$\Delta T_{\text{kettle}} = 80.0^{\circ}\text{C}$$

$$m_{\text{water}} = 1.00\text{ kg}$$

$$c_{\text{water}} = 4.18 \times 10^3\text{ J kg}^{-1}\text{C}^{-1}$$

$$\Delta T_{\text{water}} = 80.0^{\circ}\text{C}$$

$$L_v = 2.26 \times 10^6\text{ J kg}^{-1}$$

$$m_s = ?$$

$$Q = 2.00 \times 10^3\text{ J s}^{-1}$$

$$\text{Ht supplied} = \text{Ht of kettle} + \text{Ht of water}$$

+ H_f of Vap'n

$$Q = m c \Delta T_k + m c \Delta T_w + m L_v$$

$$10.0 \times 60.0 \times 2.00 \times 10^3 =$$

$$(3.00 \times 400 \times 80.0) + (1 \times 4180 \times 80.0) +$$

$$m \times 2.26 \times 10^6$$

$$m = \frac{0.770 \times 10^6}{2.26 \times 10^6}$$

$$= 0.34 \text{ kg}$$

Mass of water vaporised = $3.40 \times 10^{-1} \text{ kg}$

Mass of water left in kettle = $6.60 \times 10^{-1} \text{ kg}$

19. No. The breeze does assist in the rate at which her sweat evaporates from her body. As the sweat evaporates, the latent heat of vaporisation required is supplied by Kerry's body. This loss of heat results in the cooling sensation that she experiences.

20. Ice. As the ice melts it requires a large amount of heat (latent heat of fusion) which is additional to the heat gained by the water at 0°C in the mixture. As a consequence the soft drink will cool more effectively with the ice as most of the heat supplied to the ice will come from the soft drink.

21. $m = 0.050 \text{ kg}$
 $L_f = 3.36 \times 10^5 \text{ J kg}^{-1}$
 $Q = ?$
 $Q = mL$
 $= 0.050 \times 3.36 \times 10^5$
 $= 1.68 \times 10^4 \text{ J}$

22. $m_{\text{ice}} = 0.050 \text{ kg}$
 $L_f = 3.36 \times 10^5 \text{ J kg}^{-1}$
 $c_w = 4.18 \times 10^3 \text{ J kg}^{-1} \text{ }^\circ\text{C}^{-1}$
 $m_w = 0.300 \text{ kg}$
 $T_1 = 20.0^\circ\text{C}$
 $T_2 = T$
 Heat gained by ice = Heat lost by water
 $m L_f + m c \Delta T = m c \Delta T$
 $= (0.050)(3.36 \times 10^5) + (0.050)$
 $(4.18 \times 10^3)(T - 0)$
 $= (0.300)(4.18 \times 10^3)(20.0 - T)$
 $1.68 \times 10^4 + 209T = 25080 - 1254T$
 $1463T = 8280$
 $T = 5.7^\circ\text{C}$
 $\therefore$ Final temperature of water = 5.7°C

23. Steam at 100°C condensing to water at 100°C on the arm releases the latent heat of vaporisation acquired during the formation of the steam. It is this heat which is responsible for producing the more severe burn.

24. Liquid sodium is used as a coolant because it is a good conductor and very effective at removing heat from the reactor.

25. Gases: Collisions between gas molecules at different temperatures result in the transfer of heat from high temperature molecules to low temperature molecules. This continues until all molecules are (on average) equally energetic.

Liquids: The molecules which are at a high temperature transfer heat due to their more energetic movements by collision with cooler molecules. Heat is transferred to cooler regions of the liquid by this process.

Metals: Electrons which are free to move about the metal lattice are able to rapidly transfer energy to the cooler regions of the metal by collisions with other electrons and metal atoms.

26. Water and glass are poor conductors of heat. The transfer of heat by conduction to the ice from through the glass and water occurs slowly.

27. By fluffing its feathers the bird traps air around it beneath the outer layer of feathers. The pockets of trapped air reduce heat loss from the bird's body because air is a poor conductor of heat.

28. Pockets of air trapped between the fibres of the coat reduce heat loss from our body by conduction because both air and the fibres are poor conductors of heat.

29. Both the bricks and the insulating material have a low thermal conductivity. The insulating material slows down the rate of heat loss by conduction.

30. Conduction involves the transfer of heat as molecules of the liquid collide with cooler molecules in the liquid. Convection is the transfer of heat through the liquid by the liquid's heated molecules actually moving through the liquid. The heated portion of the liquid expands, becomes less dense than the surrounding liquid and so rises through the liquid.

31. The heat can be transferred from the fins by radiation and convection. The vertical fins can cool by convection by allowing the hot surrounding air to rise as a convection current.

32. Conduction: The flask consists of a double glass wall which contains a vacuum preventing conduction. The cork/plastic stopper is made of a material which is a poor conductor of heat.

Convection: The vacuum prevents convection. When the stopper is removed, heat can be lost by convection through the top of the flask.

Radiation: The glass walls of the flask are silvered to reduce radiation and reflect any radiated heat back into the flask.

33. Radiation. Most of the heated air rises, through convection, up the chimney.

34. Joan's woollen clothing reduced the transfer of body heat to the outside cold air, enabling her to remain warm. The low temperature around Peta's arms and legs resulted in heat loss from her body to the surroundings. Peta's body shivered to increase her metabolism so as to produce more heat to replace that which was lost to the surroundings.

35. The efficient functioning of a refrigerator relies on the heat extracted from within the refrigerator to be released outside of the refrigerator – from the coils at the back of the refrigerator. Heat dissipated from the coils must be dispersed (mainly by convection) from the coils if the refrigerator is to be efficient. An air gap around the refrigerator enables the convection to occur.

36. The intensity is less due to the effect of the atmosphere and the angle of the sun. The atmosphere always absorbs and reflects some of the solar radiation. At a latitude of 32° the radiation from the sun is spread over a larger area and this further reduces the intensity.

37. (a) The greenhouse effect is caused by gases in the atmosphere trapping heat being radiated by the Earth. This allows most of the heat that would have otherwise being lost to space to heat up the atmosphere and also be partially returned to Earth.

(b) The gases mainly responsible for the greenhouse effect are water vapour (H_2O) and carbon dioxide (CO_2). Methane (CH_4), ozone (O_3), chlorofluorocarbons (CFC's) and nitrous oxide (N_2O) also contribute.

(c) An increase in greenhouse gases would most likely contribute to global warming and cause significant changes to our environment.

38. **Greenhouse effect:** Radiation from the sun warms the earth's surface which radiates some of this heat back into space. However increased levels of carbon dioxide in the atmosphere (from the burning of fossil fuels eg coal) prevents some heat loss to space causing the atmospheric temperature to increase.

Thermal pollution: Water used as a coolant in the power station may be pumped as hot water into surrounding rivers and lakes causing a rise in temperature which can be detrimental to aquatic life.

39. (a) $P = 60\text{ W} = 60\text{ J s}^{-1}$

$3\% = 60 \times 3/100 = 1.8\text{ J s}^{-1}$

(b) The remainder of the energy is converted to heat. This warms the light bulb and fittings and is dissipated into the surroundings.

(c) Fluoro 5 times more efficient $(15/3) = 5$
Wattage needed $= 60/5 = 12\text{ W}$

40. Air high on the mountains gets very cold, contracts and becomes more dense. This cold air then flows down the mountain slopes into the valleys.

CHAPTER 4: ELECTRICITY

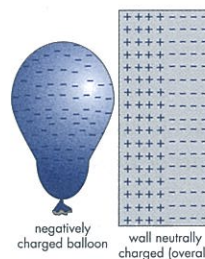
4. Chapter Questions

4.1.

Protons are located in the nucleus of atoms and are not free to move.

4.2.

The charged balloon (say negatively charged) induces an opposite charge on the surface of the wall it is near. Attraction between opposite charges results.



4.3.

$$No(e^-) = \frac{1}{1.6 \times 10^{-19}} = 6.25 \times 10^{18}$$

4.4.

$$P = \frac{W}{t} = \frac{qV}{t} = \left(\frac{q}{t}\right)(V)$$

$$\therefore P = VI$$

4.5.

As temperature rises atoms vibrate more strongly. This inhibits the flow of charge from atom to atom.

4.6.

(a) (i) At 1.5 V $I \approx 18\text{ A}$

$$\therefore R = \frac{V}{I} \approx \frac{1.5}{18} \approx 8.3 \times 10^{-2}\ \Omega$$

(ii) At 3.0 V $I \approx 23\text{ A}$

$$\therefore R = \frac{V}{I} \approx \frac{1.5}{23} \approx 6.5 \times 10^{-2}\ \Omega$$

(b)

$$P = VI = (3)(23) = 69\text{ W}$$

4.7.

(a) C would conduct most current. The greater the cross-sectional area the smaller the resistance.

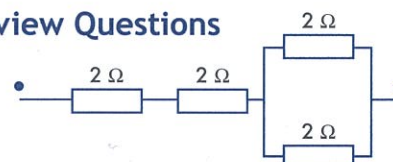
(b) B would conduct least current. Resistance increases with length.

4.8.

Parallel circuits offer several paths for current to flow. Hence resistance is always lowered when resistors are placed in parallel.

4.9.

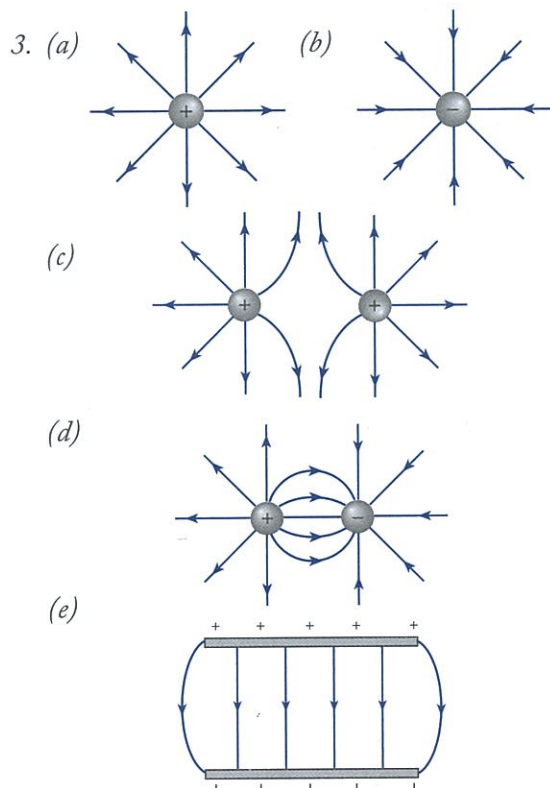
3. Review Questions



4.1 Electricity

1. No. Some electrons from the surface atoms of the woollen cloth are transferred to the rod leaving the cloth with an equal amount of positive charge. The negative charge is not created and a conservation of charge occurs between the rod and the woollen cloth.

2. A metal rod is a good conductor and any charge on the rod will quickly flow along the rod and through the body to the Earth. An insulator does not have 'free' charge carriers (electrons) to enable the charge to flow throughout the insulator.



4. Electric current is the rate of flow of electric charge past a point in a conductor and may be conventional current or electron current. Conventional current refers to the direction of current flow round a conducting circuit and is shown as the direction in which a flow of positive charge moves (ie. from a positively charged point to a negatively charged point).

$$\begin{aligned} 5. \quad q &= I t \\ &= 3.50 \times (30.0 \times 60) \\ &= 6.30 \times 10^3 \text{ C} \end{aligned}$$

$$\begin{aligned} 6. \quad q &= I t \\ 200 &= I (3.00 \times 60) \\ I &= \frac{200}{3.00 \times 60} = 1.11 \text{ A} \\ \therefore \text{Current in the lamp} &= 1.11 \text{ A} \end{aligned}$$

$$7. \quad (a) \quad P = VI$$

$$\begin{aligned} 2.40 \times 10^3 &= V \times 10 \\ \therefore V &= 240 \text{ V} \end{aligned}$$

$$\begin{aligned} (b) \quad P &= \frac{\text{Work}}{\text{time}} \\ \text{Work} &= P \times \text{time} \\ &= 2.4 \times 10^3 \times 3600 \\ &= 8.64 \times 10^6 \text{ J} \end{aligned}$$

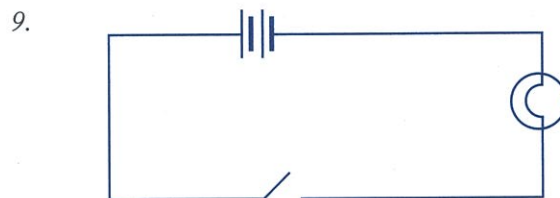
$$\therefore \text{Amount of heat energy} = 8.64 \times 10^6 \text{ J}$$

$$\begin{aligned} (c) \quad q &= I t \quad \text{OR} \quad W = V q \\ &= 10 \times 3600 \quad (8.64) \times 10^6 = 240 q \\ &= 3.60 \times 10^4 \text{ C} \quad \therefore q = 3.60 \times 10^4 \text{ C} \end{aligned}$$

$$\begin{aligned} 8. \quad (a) \quad P &= VI \\ 2.50 &= 4.50 I \\ \therefore I &= 0.556 \text{ A} \\ (b) \quad q &= I t \\ &= 0.556 \times 30.0 \\ &= 16.7 \text{ C} \end{aligned}$$

$$\begin{aligned} (c) \quad W &= V q \quad \text{OR} \quad W = P t \\ &= 4.50 \times 16.7 \quad = 2.50 \times 30.0 \\ &= 7.50 \times 10^1 \text{ J} \quad = 7.50 \times 10^1 \text{ J} \end{aligned}$$

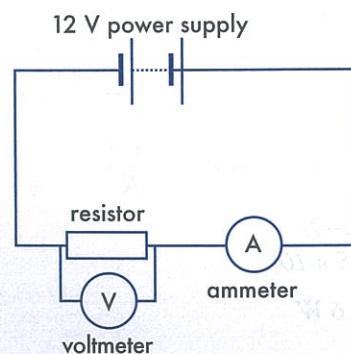
4.2 Electrical Circuits

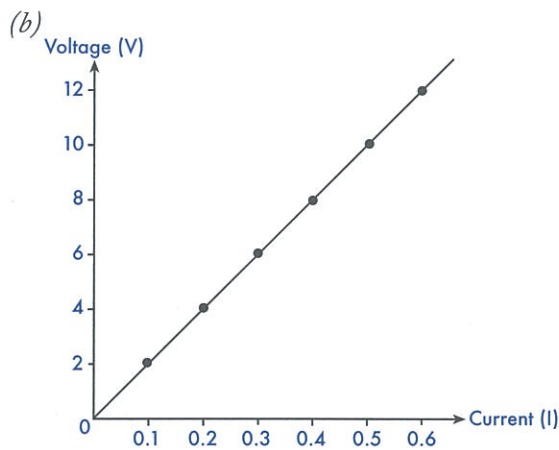


10. The current through a conductor is proportional to the potential difference across it provided its temperature remains constant.

$$\begin{aligned} 11. \quad (a) \quad P &= VI \\ 100 &= 12.0 I \\ \therefore I &= 8.33 \text{ A} \\ (b) \quad V &= IR \\ 12.0 &= 8.33 R \\ \therefore R &= 1.44 \Omega \end{aligned}$$

12. (a)





Resistance = gradient

$$= \frac{10 - 4}{0.5 - 0.2}$$

$$= \frac{6}{0.3}$$

$$= 20$$

$\therefore$ Resistance = 20 ohm

13.

(a) $V = IR$
 $3.00 = 0.300 R$
 $\therefore R = 10.0 \Omega$

(b) $P = VI$
 $= 3.00 \times 0.300$
 $\therefore P = 0.900 W$

14.

(a) Potential difference across each globe
 $= \frac{240}{16} = 15 V$

(b) Power = $\frac{V^2}{R}$
 $R = \frac{(240)^2}{24} = 2.40 \times 10^3 \Omega$

Resistance of circuit = $2.40 \times 10^3 \Omega$

$\therefore$ Resistance of each globe = $\frac{2.40 \times 10^3}{16}$

= 150 ohm

15. Total resistance = 15 globes $\times$ 150 ohm
 $= 2.25 \times 10^3 \text{ ohm}$

Power consumption = $\frac{V^2}{R}$

= $\frac{(240)^2}{2.25 \times 10^3}$

= 25.6 W

16. Maximum power output = VI

= 240×10

= 2400 W

$\therefore$ Number of 75 W light globes
 which would blow the fuse = $\frac{2400}{75}$

= 32

$\therefore$ Maximum number of light globes to
 be used

= 31

17.

(a) $R_T = R_1 + R_2$
 $= 11 + 7$
 $= 18 \Omega$

$\therefore$ Resistance of circuit = 18 Ω

(b)

$V = IR$

$9.0 = I \times 18$

$\therefore I = 0.50 A$

$\therefore$ Current through circuit = 0.50 A

18.

(a) $\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2}$
 $= \frac{1}{18} + \frac{1}{6}$

= $\frac{1+3}{18}$

= $\frac{4}{18}$

$\therefore R_T = 4.5 \Omega$

$\therefore$ Resistance of circuit = 4.5 A

(b)

$V = IR$

$9.0 = I \times 4.5$

$\therefore I = 2.0 A$

(c)

$I_1 = \frac{9.0}{1.8}$

= 0.5 A

(d)

$I = I_1 + I_2$

$2.0 = 0.5 + I_2$

$\therefore I_2 = 1.5 A$

$\therefore PD = IR$

= 1.5×6.0

= 9.0 V

19.

(a) In parallel section: $\frac{1}{R_T} = \frac{1}{48} + \frac{1}{24}$

= $\frac{1+2}{48} = \frac{3}{48}$

$\therefore R_T = 16 \Omega$

$\therefore$ Total resistance of circuit = 8.0 + 16

= 24 Ω

$$(b) \quad V = IR$$

$$6.0 = I \times 24$$

$$\therefore I = 0.25 \text{ A}$$

$$(c) \quad \text{P.D. across parallel section} = IR_T$$

$$= 0.25 \times 16$$

$$= 4.0 \text{ V}$$

$$\therefore \text{P.D. across } 24 \Omega \text{ resistor} = 4.0 \text{ V}$$

$$(d) \quad P = I^2 R$$

$$= (0.25)^2 \times 8.0$$

$$= 0.50 \text{ W}$$

$$\therefore \text{Power dissipated through } 8.0 \Omega \text{ resistor} = 0.50 \text{ W}$$

20.

$$(a) \quad V = IR$$

$$= 0.50 \times 20$$

$$= 10 \text{ V}$$

$$(b) \quad \text{P.D. across the rheostat} = 12 - 10 = 2.0 \text{ V}$$

$$\therefore V = IR$$

$$2.0 = 0.50 R$$

$$R = 4.0 \Omega$$

$$\therefore \text{Resistance of the rheostat} = 4.0 \Omega$$

21.

Electrical energy from the power source is converted to internal energy in and released as heat from the coil. The potential difference across the coil produces a current in the coil which in turn releases energy in the form of heat. This heat raises the temperature of the water.

22.

$$(a) \quad \text{Electrical energy supplied} = P \times t$$

$$= 2400 \times 50.0 = 1.20 \times 10^5 \text{ J}$$

(b) No. Some of the internal energy transformed from the electrical energy in the kettle will be used to heat the kettle and escape to the surroundings.

23. A fuse is a short thin piece of wire which has a low melting point. When an electric current which is larger than what it is designed to carry passes through the fuse, it heats up, melts and breaks the circuit. This prevents the larger currents from flowing into and damaging electrical appliances or from causing a fire due to an appliance overheating.

24.

$$(a) \quad P = VI$$

$$180 = 240 I$$

$$\therefore I = 0.75 \text{ A}$$

$$\therefore \text{Use a } 2.00 \text{ A fuse.}$$

(b) Components in the television set may be damaged by the excessive current or a fire may start in the television set before the 10.0 A fuse is made to 'blow' by this larger current.

25. An earth leakage circuit breaker can:

- (i) switch off an electric current in a very short time,
- (ii) quickly stop an electric shock or reduce the chances of an electrocution occurring,
- (iii) prevent a fire occurring in an electrical appliance due to a fault causing a current to flow through earth wires.

26. Wires may be damaged in the accident or by moving the damaged vehicle causing short circuits. The large currents supplied by the car's battery can cause leads which are short circuited to overheat and start a fire. Also, hot wires and sparks could ignite spilt petrol.

27. Parallel - so that either of them can supply an electric current to the car's circuits in the event that one breaks down while the car is in use.

28.

$$(a) \quad P = VI = 12.0 \times 3.50 \times 10^2 = 4.20 \times 10^3 \text{ W}$$

$$(b) \quad R = \frac{V}{I} = \frac{12.0}{3.50 \times 10^2} = 3.43 \times 10^{-2} \Omega$$

(c) The battery to starter motor wire is much thicker for it to have a small resistance (less than $3.43 \times 10^{-2} \Omega$) and to carry a very large current.